21/02/25

Lecture 14

Virial expansion of Eq^n of State

If we consider a dilute gas, then we can define the cluster cofficients at high volume limit, where

$$\bar{b}_l(T) = \lim_{V \to \infty} b_l(V, T)$$

The virial expansion of eq^n of state is defined as,

$$\frac{Pv}{k_B T} = \sum_{l=1}^{\infty} a_l \left(\frac{\lambda^3}{v}\right)^{l-1}$$

$a_l \Rightarrow l^{th}$ virial coffecient.

$$\frac{Pv}{k_B T} = a_1 + a_2\left(\frac{\lambda^3}{v}\right) + a_3\left(\frac{\lambda^6}{v^2}\right) + \dots \text{ so on.}$$

At high volume limit (for dilute gas) the cluster expansion becomes,

$$\frac{P}{k_B T} = \frac{1}{\lambda^3} \sum_{l=1}^{\infty} \bar{b}_l z^l$$

& $$\frac{1}{v} = \frac{1}{\lambda^3}\sum_{l=1}^{\infty} l b_l z^l$$

$$\Rightarrow \frac{\left(\frac{P}{k_B T}\right)}{(1/v)} = \frac{Pv}{k_B T} = \frac{\sum_{l=1}^{\infty} b_l z^l}{\sum_{l=1}^{\infty} l b_l z^l}$$

& put the virial expansion of

$\frac{Pv}{k_B T} = \sum_{l=1}^{\infty} a_l \left(\frac{\lambda^3}{v}\right)^{l-1}$ in the above eq^n.

$$\Rightarrow \sum_{l=1}^{\infty} a_l \left[\frac{\lambda^3}{v}\right]^{l-1} = \frac{\sum_{l=1}^{\infty} b_l z^l}{\sum_{l=1}^{\infty} l b_l z^l}$$

Put $\frac{\lambda^3}{v} = \sum_{l=0}^{\infty} l b_l z^l$ above

$$\Rightarrow \sum_{l=1}^{\infty} a_l \left[\sum_{n=0}^{\infty} n b_n z^n\right]^{l-1} = \frac{\sum_{l=1}^{\infty} b_l z^l}{\sum_{l=1}^{\infty} l b_l z^l}$$

$$\Rightarrow \left[\sum_{l=1}^{\infty} l b_l z^l\right]\left[\sum_{l=1}^{\infty} a_l \left[\sum_{n=0}^{\infty} n b_n z^n\right]^{l-1}\right] = \sum_{l=1}^{\infty} b_l z^l$$

The relations b/w a_ℓ & b_ℓ can be obtained by equating the power of "z" in both the sides of eq^n by requiring that the equality holds for any value of z.

$$\left[b_1 z + 2b_2 z^2 + 3b_3 z^3 + \cdots\right]$$
$$\times \left\{a_1 + a_2\left[\sum_{n=0}^{\infty} n b_n z^n\right] + a_3\left[\sum_{n=0}^{\infty} n b_n z^n\right]^2 + \cdots\right\}$$
$$= b_1 z + b_2 z^2 + b_3 z^3 + \cdots$$

By equating the powers of z we get the following relations.

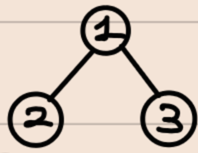
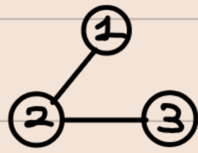
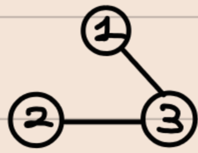
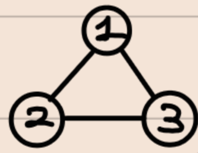
$a_1 = b_1 \qquad a_2 = -b_2 \qquad a_3 = 4b_2^2 - 2b_3$

$$a_4 = -20 b_2^3 + 18 b_2 b_3 - 3b_4$$

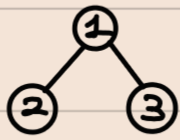
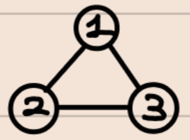
$$a_5 = 112 b_2^4 - 144 b_2^2 b_3 + 18 b_3^2 + 32 b_2 b_4 - 4 b_5$$

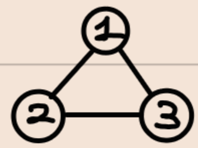
$$a_6 = -672 b_2^5 + 1120 b_2^{(3)} b_3 - 315 b_2 b_3^3$$
$$- 280 b_2^2 b_4 + 60 b_3 b_4 + 50 b_2 b_5 - 5 b_6$$

Consider the definition of 'b_3'

$b_3 = \frac{1}{3!\,\lambda^6 V}\Big[$  $+$  $+$  $+$  $\Big]$

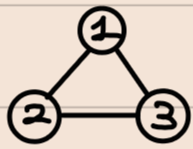
Numerically equal to $\int d^3r_1 d^3r_2 d^3r_3\, f_{12} f_{13}$

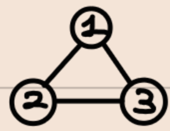
$\Rightarrow b_3 = \frac{1}{3!\lambda^6 V}\Big[3$  $+$  $\Big]$

$\Rightarrow b_3 = \frac{1}{3!\lambda^6 V}\Big[3\int d^3r_1 d^3r_2 d^3r_3\, f_{12} f_{13} +$  $\Big]$

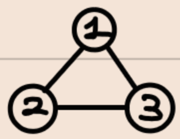
$r_1 \to r_1 - r_3 \qquad r_2 \to r_1 - r_2$

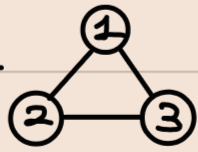
$$= \frac{1}{2\lambda^6 V}\Big[\underbrace{\int d^3r_3}_{V}\int d^3r_{13}\, d^3r_{12}\, f_{12}(r_{12})\, f_{13}(r_{13})\Big]$$

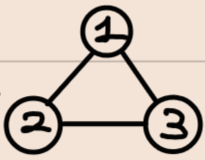
$+ \frac{1}{3!\lambda^6 V}$ 

$= \frac{1}{2\lambda^6}\underbrace{\Big[\int d^3r_{12}\, f_{12}(r_{12})\Big]}_{2\lambda^3 b_2}\underbrace{\Big[\int d^3r_{13}\, f(r_{13})\Big]}_{2\lambda^3 b_2} + \frac{1}{3!\lambda^6 V}$ 

$$\Big[b_2 = \frac{1}{2\lambda^3}\int d^3r_{12}\, f_{12}(\vec{r}_{12})\Big]$$

$\Rightarrow b_3 = 2b_2^2 + \frac{1}{3!\lambda^6 V}$ 

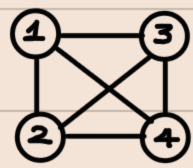
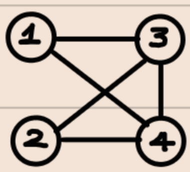
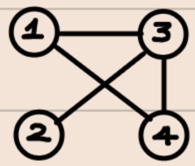
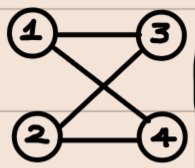
$$\Rightarrow \underbrace{2b_3 - 4b_2^2}_{-a_3} = \frac{1}{3\lambda^6 V}$$ 

$$\Longrightarrow \boxed{a_3 = \frac{-1}{3\lambda^6 V}}$$ 

Using the relations b/w a_l & b_l we can actually show that the l^{th} virial coefficient is proportional to the sum of all the irreducible l-clusters.

$$\underset{(l>2)}{a_l} = -\frac{\text{Sum of all irreducible } l\text{-clusters}}{l\,(l-2)!\;\lambda^{3(l-1)}\,V}$$

Here irreducible means multiply connected, i.e. if there exists more than one path connecting any two vertices. Eg:

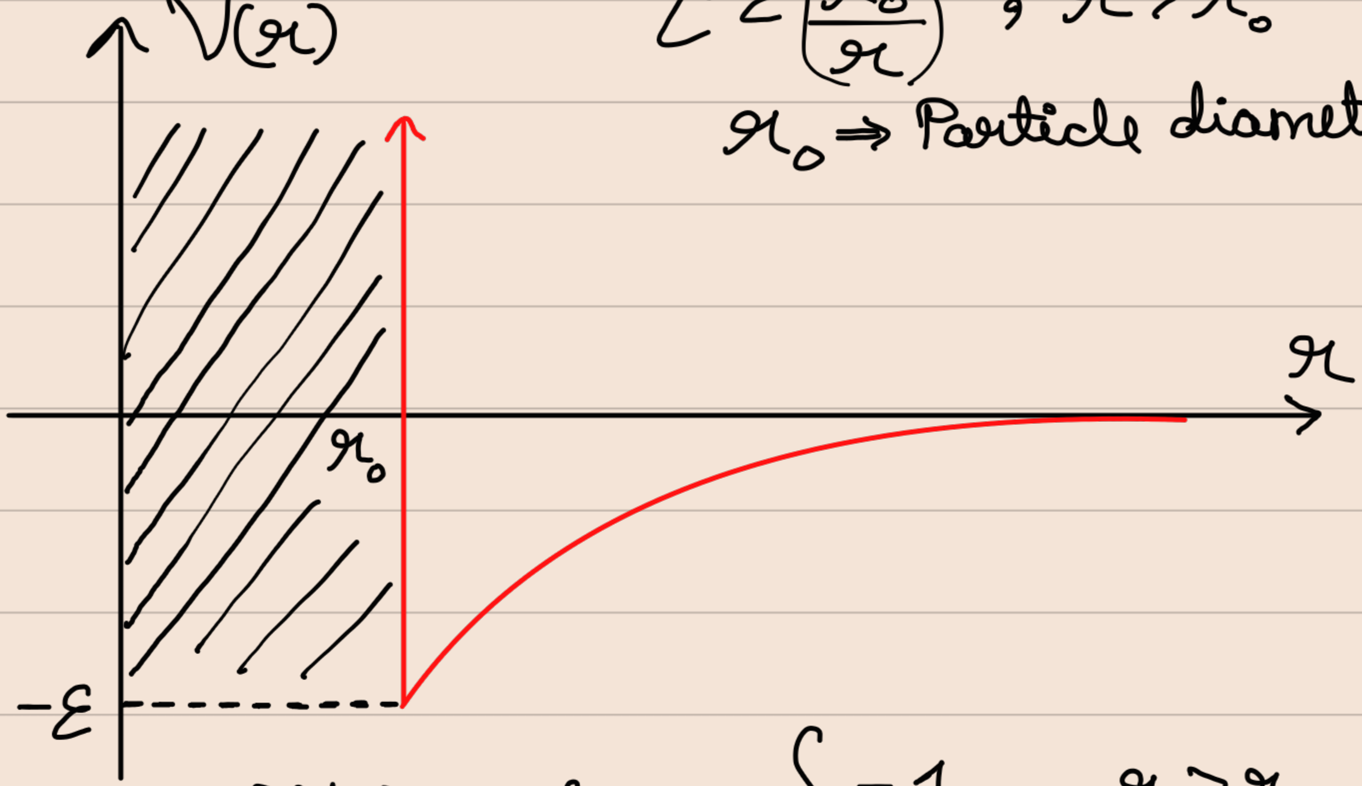
$$a_4 = \frac{-1}{8\lambda^9 V}\Big[$$  $+6$  $+12$  $+3$  $\Big]$

Vander waal's gas

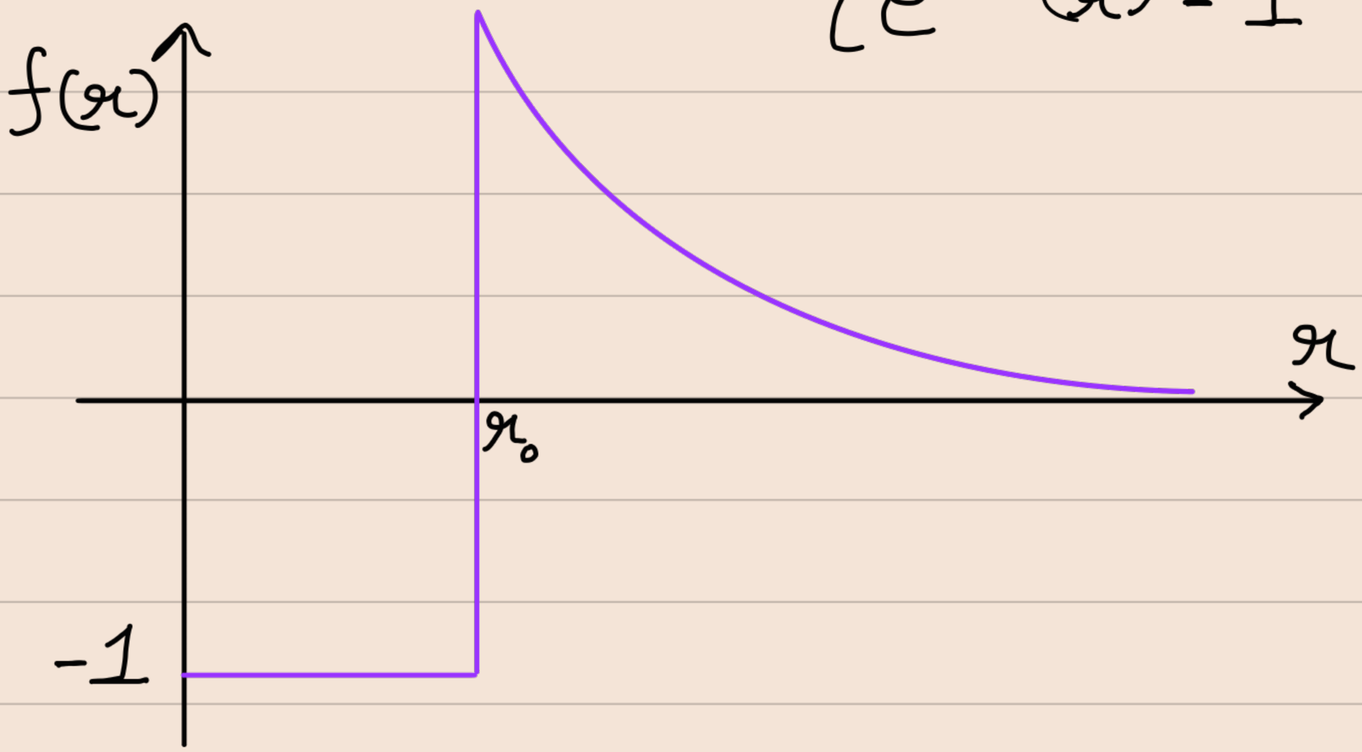
The pair interaction is of the form

$$V(r) = \begin{cases} \infty , & r < r_0 \\ -\varepsilon \left(\frac{r_0}{r}\right)^6 , & r > r_0 \end{cases}$$

$r_0 \Rightarrow$ Particle diameter



$$e^{-\beta V(r)} - 1 = f(r) = \begin{cases} -1 , & r > r_0 \\ e^{\beta \varepsilon \left(\frac{r_0}{r}\right)^6} - 1 \end{cases}$$



$$b_2 = \frac{1}{2\lambda^3} \int_{\text{all space}} d^3 r \, f(r)$$

$$= \frac{1}{2\lambda^3} \int_0^\infty f(r) \, 4\pi r^2 \, dr$$

$$= \frac{2\pi}{\lambda^3} \left[\int_0^{r_0} (-1) r^2 dr + \int_{r_0}^\infty \left(e^{\frac{\beta \varepsilon r_0^6}{r^6}} - 1 \right) r^2 dr \right]$$

$$= \frac{-2\pi r_0^3}{3\lambda^3} + \frac{2\pi}{\lambda^3} \int_{r_0}^\infty \left(e^{\frac{\beta \varepsilon r_0^6}{r^6}} - 1 \right) r^2 dr$$

Let $t = \frac{r_0}{r}$ $\qquad dt = \frac{-r_0 \, dr}{r^2}$

$$b_2 = \frac{-2\pi r_0^3}{3\lambda^3} + \frac{2\pi r_0^3}{\lambda^3} \int_0^1 dt \frac{\left[e^{\beta \varepsilon t^6} - 1 \right]}{t^4}$$

High temperature expansion will simplify the calculation.

$$e^{\beta\varepsilon t^6} - 1 = \beta\varepsilon t^6$$

$$\Rightarrow b_2 = \frac{-2\pi r_0^3}{3\lambda^3}\left[1 - 3\int_0^1 \frac{\beta\varepsilon t^6\,dt}{t^4}\right]$$

$$b_2 = \frac{-2\pi r_0^3}{3\lambda^3}\left[1 - \beta\varepsilon\right]$$

$$\bar{b}_2 = b_2 \quad \text{(as this is } V \text{ independent)}$$

$$\Rightarrow a_2 = -\bar{b}_2 = \frac{2\pi r_0^3}{3\lambda^3}(1 - \beta\varepsilon)$$

The virial expansion of eq^n of state is given by

$$\frac{Pv}{K_B T} = 1 + a_2\left(\frac{\lambda^3}{v}\right) \qquad \left(\text{upto } 2^{nd} \text{ order}\right)$$

$$\Rightarrow \frac{Pv}{K_B T} = 1 + \frac{2\pi r_0^3}{3v}(1 - \beta\varepsilon)$$

Let $b = \frac{2\pi}{3} r_0^3$. Since r_0 is the diameter of particles, $b = 4 \times$ molecular volume.

$$\Rightarrow \frac{Pv}{k_B T} = 1 + \frac{b}{v} - \frac{\beta b \varepsilon}{v}$$

$$\Rightarrow \left[\frac{Pv}{k_B T} + \frac{\beta b \varepsilon}{v}\right] = \left[1 + \frac{b}{v}\right]$$

$$\Rightarrow \left[P + \frac{b\varepsilon}{v^2}\right] = \frac{k_B T}{v}\left[1 + \frac{b}{v}\right]$$

If $\frac{b}{v}$ is very small, then

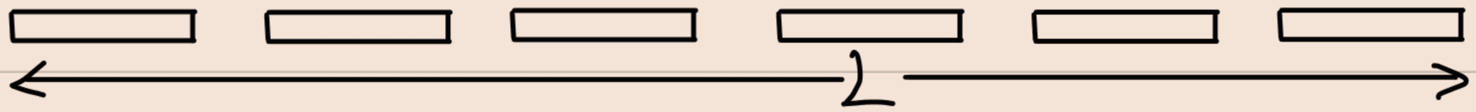
$$\left(1 + \frac{b}{v}\right) \sim \frac{1}{\left(1 - \frac{b}{v}\right)}$$

$$\Rightarrow \left[P + \frac{b\varepsilon}{v^2}\right] \approx \frac{k_B T}{[v - b]}$$

$$\Rightarrow \boxed{\left[P + \frac{b\varepsilon}{v^2}\right]\left[v - b\right] = k_B T}$$

↳ Vander waal's eqn of state

Tonk's Gas (Rods in a 1 D Container)

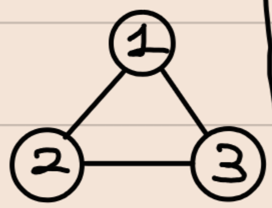


$$V_{ij}(|x_i - x_j|) = \begin{cases} \infty \,, & |x_i - x_j| \leq a \\ 0 \,, & |x_i - x_j| > a \end{cases}$$

($L \Rightarrow$ length of Container (1-D volume))

$$a_1 = 1$$

$$a_2 = \frac{-1}{2\lambda L}\Bigg[$$  $$\Bigg]$$

$$a_3 = \frac{-1}{3\lambda^2 L}\Bigg[$$  $$\Bigg]$$

For 1D a_1, a_2 & a_3 are virial coefficients.

when $x < a$, $f(x) = -1$

when $x > a$, $f(x) = 0$

$$\Rightarrow a_2 = \frac{-1}{2\lambda L}\int_0^L dx_1\, dx_2\, f_{12}(x_1 - x_2)$$

$$= \frac{-1}{2\lambda}\int_{-L}^{L} dx\, f_{12}(x) = \frac{-1}{\lambda}\int_0^L dx\, f_{12}(x)$$

$$\Rightarrow \boxed{a_2 = \frac{a}{\lambda}}$$

$$a_3 = \frac{-1}{3L\lambda^2}\int_0^L\int_0^L\int_0^L dx_1 dx_2 dx_3\, f_{12} f_{13} f_{23}$$

By translational invariance, set x_1 at origin. & vary x_2 & x_3

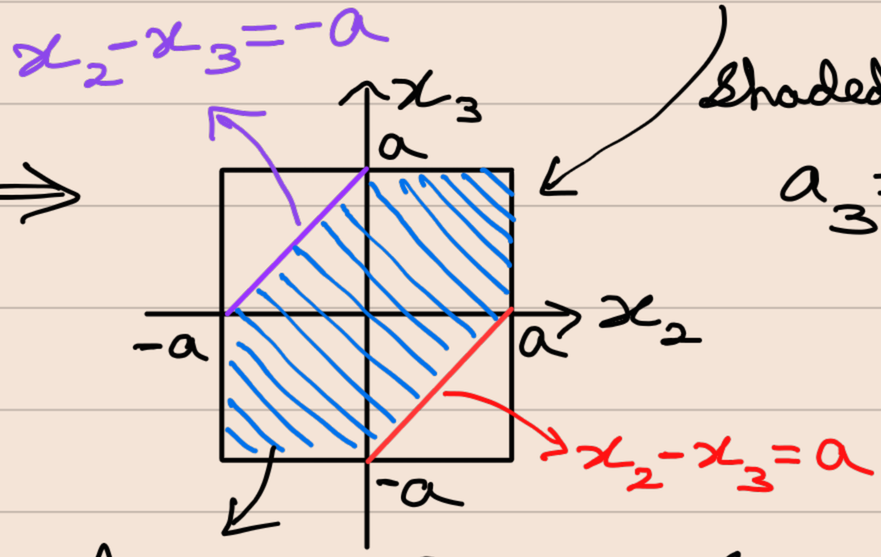
$$\rightarrow a_3 = \frac{-1}{3\lambda^2}\int_0^L\int_0^L dx_2 dx_3\, f(0,x_2)\, f(x_2,x_3)\, f(0,x_3)$$

$$f(0,x_2) = -1 \quad \text{when} \quad |x_2| \le a$$
$$f(0,x_3) = -1 \quad \text{when} \quad |x_3| \le a$$
$$\& \; f(x_2,x_3) = -1 \quad \text{when} \quad |x_2 - x_3| \le a$$

$$\Rightarrow a_3 = \frac{-1}{3\lambda^2}\int_{-a}^{a} dx_2 \underbrace{\int_{-a}^{a} dx_3\, f(|x_2 - x_3|)}$$

Shaded area

$\Rightarrow$



Area $= 3a^2$

$$a_3 = \frac{-1}{3\lambda^2}(-\text{Area})$$

$$\boxed{a_3 = \frac{a^2}{\lambda^2}}$$

Similarly $a_n = \left(\frac{a}{\lambda}\right)^{n-1}$